

Asymptotics and inverses for a stable Hilbert series

The coefficients of OEIS A126348

1 October 2026

Abstract

We derive a relative all-orders expansion for the coefficients of $\prod_{k \geq 1} (1 + q^k / (1 - q))$, a stable Hilbert series associated with top-degree Grothendieck and Lascoux polynomials. A quadratic logarithmic saddle provides a resummed leading term and rational correction functions, with a finite rule for every order and a uniform remainder. An exact Jacobi and eta factorization separately identifies convergent oscillatory sectors of the generating function. The coefficient theorem yields an explicit inverse with $o(1)$ error in the index on the sequence range, and qualified integer threshold rules.

1 The coefficient theorem

Define

$$F(q) = \prod_{k \geq 1} \left(1 + \frac{q^k}{1 - q} \right) = \sum_{n \geq 0} a_n q^n, \quad |q| < 1. \quad (1)$$

The sequence begins 1, 1, 2, 4, 7, 12, 20, 33, 53, 84, 131 and is A126348 [1]. It counts colored partitions with distinct colors [3]. Pan and Yu identify (1) as the Hilbert series of the stable span of top-degree Grothendieck and Lascoux polynomials [2, Proposition 6.10].

Put $c = \pi^2/6$ and

$$A(L) = \frac{1}{2}L^2 + L + c, \quad B(L) = L^2 + L + 2c, \quad V(L) = L^2 + 3L + 2c + 1.$$

For sufficiently large n , let $L = L_n > 0$ be the unique solution of

$$n = e^{2L} A(L), \quad t = e^{-L}. \quad (2)$$

All expansion orders are fixed while $n \rightarrow \infty$.

Theorem 1.1. *For each integer $J \geq 0$, there are rational functions $C_j \in \mathbb{Q}(c, L)$, with $C_0 = 1$, such that*

$$a_n = \frac{\exp\{e^L B(L) - 1\}}{\sqrt{2\pi e^{3L} V(L)}} \left(\sum_{j=0}^J e^{-jL} C_j(L) + O_J(e^{-(J+1)L} (1+L)^{J+1}) \right). \quad (3)$$

Moreover, $C_j(L) = O_j((1+L)^j)$. A finite exact rule for all C_j is provided in Section 5.

This is a coefficient equivalent, not only a logarithmic equivalent. Keeping (2) exact retains infinitely many corrections in inverse powers of $\log n$.

For explicit formulas define

$$V_0 = \frac{1}{2}L^2 + c, \quad V_j = (j + \partial_L) V_{j-1}. \quad (4)$$

Thus $V_1 = A$, $V_2 = V$, and

$$\begin{aligned} V_3 &= 3L^2 + 11L + 6c + 6, \\ V_4 &= 12L^2 + 50L + 24c + 35, \\ V_5 &= 60L^2 + 274L + 120c + 225, \\ V_6 &= 360L^2 + 1764L + 720c + 1624. \end{aligned}$$

Let $P_1 = (5 - L)/24$, $P_2 = -1/9$, and

$$E_1 = \frac{V_4}{8V^2} - \frac{5V_3^2}{24V^3}, \quad (5)$$

$$E_2 = -\frac{V_6}{48V^3} + \frac{7V_3V_5}{48V^4} + \frac{35V_4^2}{384V^4} - \frac{35V_3^2V_4}{64V^5} + \frac{385V_3^4}{1152V^6}. \quad (6)$$

The first two corrections are

$$C_1 = P_1 + E_1, \quad (7)$$

$$C_2 = P_2 + \frac{1}{2}P_1^2 + P_1E_1 + E_2 - \frac{1}{48V} - \frac{(6 - L)V_3}{48V^2}. \quad (8)$$

The last two terms of C_2 arise from the displaced amplitude.

1.1 Provenance, status and reading conventions

[write] **Added 5 October 2026, batch 100.** *Provenance.* This report was built from one manuscript: Report 84 of the session bundle placed as batch 100 of the repository’s incoming-reports channel, archive `ProveIt_A126348_Hilbert_Asymptotics_and_Inverses.zip` (wrapper directory `A126348_Hilbert_Asymptotics_and_Inverses/`; main file `article.tex`, 450 lines, dated 1 October 2026, with a 7-page PDF that is not shipped), arrival commit `60f54ea06`, placed in `36571ae0e`. The text from the title to the bibliography is the delivered manuscript, unchanged except for the label prefix `shs:`, this subsection, Section 9 and the other notes marked [write]. No statement, proof, number or symbol of it was changed, and no section, theorem or equation number moved. The delivered expanded derivation (`derivation.md`), the delivery’s internal analytic review (`AUDIT.md`, dated 1 October 2026) and its source and scope audit (`source_audit.md`) are shipped beside this article, byte-identical; the seven programs are in `code/` and their recorded outputs in `data/`. *Pin.* The source audit checked the ProveIt snapshot `1512ef835` (1 October 2026). The manuscript continues no repository report. *Status.* An unrefereed research manuscript from an AI-assisted delivery channel; its author line is empty. The delivery’s provenance record (not shipped) gives its status as “Analytic proof, independent coefficient reconstruction, and final article transcription reviewed; numerical onset not certified.” The review meant there is the delivery pipeline’s own (`AUDIT.md`), not external peer review. Nothing in the report is formalized in Lean or Rocq, and its placement in the repository confers no formal status. *Checks.* At intake and again at this write all seven delivered programs were rerun on a copy of the delivered layout; all sixteen regenerated outputs equal the shipped ones up to line endings. At intake the coefficients were regenerated independently through $n = 6000$ from Somos’s sum formula for A126348 (not from the product), the ratio of the residual after C_1 to $e^{-2L}C_2$ was found to tend to one (0.946, 0.959, 0.974, 0.981 at $n = 500, 1000, 3000, 6000$), and (9) was confirmed to 60 digits at $t = 0.3, 0.7, 1.5$. These are numerical diagnostics; the proofs of Sections 2–7 were checked by reading.

[write] **Dated corrections to delivered text, 5 October 2026.** (1) The delivered `source_audit.md` reports that the snapshot `1512ef835` has “no A126348, A022629, A266891, A124380 match”. That was true at the pin and is stale now: A022629 is the subject of the report `a022629-distinct-partition-norms` (placed in `9df4ba51a`), A266891 is its Section 20 (`aa7345800`), and A124380 is the subject of `a124380-signed-moment-asymptotics` (`4f11bc9c0`), all in this collection. The ranked screen of the source audit, which hands A022629, A124380 and A266891 to other work, describes sibling projects of the same session that are now placed; it raises no question for this report. A126348 itself is treated nowhere else in the repository. (2) The source audit says that the OEIS entry “defines” A126348 by the

product (1). The OEIS name is “Limit of reversed rows of triangle A126347, in which row sums equal Bell numbers (A000110)”; the product is one of its three generating functions (entry read on 5 October 2026). (3) The delivered README’s sentence “Exact coefficient generation through 10000 is included” means that the programs generate the coefficients; only eight selected values ($n = 50, \dots, 10000$) are recorded, in `data/coefficients_selected.json`.

[write] Reading conventions. The manuscript reuses several letters; no symbol was renamed. Five readings need care. *The fugacity* $1/(1-q)$ depends on q , not on the index n : (1) is a single power series, unlike the size-dependent fugacity n^α of **a291698-moving-fugacity-partitions**, and the letter u here is the theta phase $\lambda/t - 1/2$, not a fugacity (the fugacity is $1/\delta = e^\lambda$). *Q* has two meanings: the modular nome $e^{-4\pi^2/t}$ in Section 2, and the cubic-and-higher phase series of Section 5, used also in Section 4. *N* has two meanings: the smooth function $N(L) = e^{2L}A(L)$ (with N_0 and $\hat{N}$) in Section 6, and the integer threshold $N(Y) = \min\{n \geq 2 : a_n \geq Y\}$ in Section 7. *The calligraphic $\mathcal{G}$ of Section 5 is a Gaussian moment functional* (its macro is named `\E`); it is neither the partner logarithm $G(t)$ of (11) nor an expectation of E_1, E_2 . *C_1 in the proof of Lemma 3.1 is an unnamed positive constant*, not the correction function $C_1(L)$ of (7); likewise C in Section 7.

symbol	meanings, by section
A, B, V	$A(L), B(L), V(L)$ of Section 1, with V_j from (4) ($V_1 = A, V_2 = V$); B_{2m} are Bernoulli numbers (Section 3); B_1 is the next inverse coefficient (Section 6).
c, C	$c = \pi^2/6$; $c_0, \dots, c_4, c_J, C, C_1$ (in the proof of Lemma 3.1), C_J are unnamed constants; $C_j(L)$, $j \geq 0$, are the correction functions of Theorem 1.1.
D	$D_r(u)$ the sector amplitudes (14); $D = 1 - q $ in Section 4; $D_2 = C_2 - C_1^2/2$ and the operator $D_y = e^{-L}V^{-1}\partial_L$ in Section 6.
$E, \mathcal{G}, G$	E_1, E_2 of (5)–(6); $\mathcal{G}$ the Gaussian moment functional; $G(t)$ the partner logarithm (11).
H, h	$H(L) = e^L B(L)$ the inverse carrier (18), but $H(\epsilon) = \sum h_k \epsilon^k$ in the recurrence of Section 5; $h(L)$ the logarithmic forward correction of Section 6.
K	$K(t)$ the exact core exponent (10); K_0 a large fixed constant (Lemma 3.1).
L, t	$L = \log(1/t)$ with $t = e^{-L}$: the forward saddle L_n of (2) in Sections 1–5, but the root of the inverse carrier (18) in Sections 6–7. In a291698-moving-fugacity-partitions , $L = \log u$ is the logarithm of the fugacity, which is λ here.
P_j, p	$P_j(L)$ the coefficients of (15); $p(k)$ the partition function in (14).
Q, R	see above for Q ; R_J the logarithmic amplitude series of Section 5; R the sector cutoff of Theorem 2.1.
r	the sector index of (13); the radius $r = e^{-t}$ in Section 4; the summation index of R_J .
T	$T(L) = 1 + \frac{3}{2}L + \frac{1}{2}\log(2\pi V)$, not the saddle t .
u, w	$u = \lambda/t - 1/2$ the theta phase; $w_k = r^k/(1 - r)$ in the minor-arc estimate, but $w = -i\epsilon X/\sqrt{V}$ in the central arc and in Section 5.
δ, λ	$\delta = 1 - q$, $\lambda = -\log \delta$; at complex z , $\delta(z) = 1 - e^{-z}$ and $\lambda(z) = -\text{Log } \delta(z)$.

2 An exact modular factorization

For real $t > 0$ set

$$q = e^{-t}, \quad \delta = 1 - q, \quad \lambda = -\log \delta, \quad u = \lambda/t - 1/2, \quad Q = e^{-4\pi^2/t}.$$

Use $(z; q)_\infty = \prod_{k \geq 0} (1 - zq^k)$.

Theorem 2.1. *The following identity is exact:*

$$F(e^{-t}) = e^{K(t)} \frac{\Theta(u, t)}{(Q; Q)_\infty}, \quad (9)$$

$$K(t) = \frac{\lambda^2/2 + c}{t} - \frac{\lambda}{2} + \frac{t}{12} - G(t), \quad (10)$$

$$G(t) = \log(-\delta; e^{-t})_\infty = \sum_{j \geq 1} \frac{(-1)^{j+1} \delta^j}{j(1 - e^{-jt})}, \quad (11)$$

$$\Theta(u, t) = \sum_{m \in \mathbb{Z}} e^{-2\pi^2 m^2/t} e^{2\pi i m u}. \quad (12)$$

Writing $p(k)$ for the ordinary partition function, with $p(0) = 1$, the normalized quotient has the exact convergent expansion

$$\frac{\Theta(u, t)}{(Q; Q)_\infty} = \sum_{r \geq 0} D_r(u) e^{-2\pi^2 r/t}, \quad (13)$$

$$D_r(u) = \sum_{\substack{m \in \mathbb{Z}, m^2 \leq r \\ r-m^2 \text{ even}}} p\left(\frac{r-m^2}{2}\right) e^{2\pi i m u}. \quad (14)$$

For fixed R , the remainder after $r = R$ is $O_R(e^{-2\pi^2(R+1)/t})$ as $t \downarrow 0$, uniformly in real u .

Proof. Jacobi's triple product [4, 17.8.1] gives

$$F(q)(-\delta; q)_\infty (q; q)_\infty = \sum_{k \in \mathbb{Z}} e^{-tk^2/2 + (\lambda - t/2)k}.$$

Completing the square and applying Poisson summation turns the right side into

$$\sqrt{2\pi/t} e^{(\lambda - t/2)^2/(2t)} \Theta(u, t).$$

The eta transformation [4, 17.2.6_1] is

$$(q; q)_\infty = \sqrt{2\pi/t} e^{-\pi^2/(6t) + t/24} (Q; Q)_\infty.$$

The square-root factors cancel and $t/8 - t/24 = t/12$. Expanding the logarithms in the partner product gives (11), absolutely convergently since $0 < \delta < 1$. Multiply the theta series by $(Q; Q)_\infty^{-1} = \sum_{k \geq 0} p(k) Q^k$ to obtain (14). Its convergence and tail follow from $p(k) = e^{O(\sqrt{k})}$. $\square$

The first amplitudes are

$$D_0 = 1, \quad D_1 = 2 \cos(2\pi u), \quad D_2 = 1, \quad D_3 = 2 \cos(2\pi u), \quad D_4 = 2 + 2 \cos(4\pi u).$$

These are exactly normalized oscillatory sectors of the generating function relative to the exact core $e^{K(t)}$. They do not by themselves identify exponentially small coefficient sectors.

[write] Credit: a special case of a291698's identity (11), 5 October 2026. The identity (9) is not new. It is the exact Jacobi identity (11) of a291698-moving-fugacity-partitions (Section 5 there, label `mfp:exact-jacobi-decomposition-and-the-finite-sector-proof`), which holds for $\prod_{k \geq 1} (1 + uq^k)$ with $q = e^{-z}$, for every fixed $u > 1$ and $\Re z > 0$, specialized pointwise to the fugacity $u = 1/\delta(t)$. For real $t > 0$ one has $0 < \delta < 1$, so this fugacity exceeds one and (1) is that product at $z = t$. With $\log(1/\delta) = \lambda$ (called L there), $u^{-1/2} = e^{-\lambda/2}$, $(-1/u; e^{-t})_\infty = (-\delta; e^{-t})_\infty = e^{G(t)}$, and $(-1)^j e^{2\pi i j \lambda/t} = e^{2\pi i j (\lambda/t - 1/2)}$, identity (11) becomes (9) term by term; the proof above (triple product,

Gaussian Poisson summation, eta transformation) is the proof given there. The manuscript does not cite that report: its pin `1512ef835` predates the report's placement (`4f11bc9c0`, 2 October 2026). At this write both sides of (11) at $u = 1/\delta(t)$, and (9), were evaluated against the product at $t = 0.3, 0.7, 1.5$ with 50 digits (relative differences below 10^{-49}). What this report adds is the use of the identity, not the identity: with a q -dependent fugacity it is an exact factorization of one power series; the sector expansion (13)–(14) (new, but elementary: the product of two absolutely convergent series); and its role in the uniform complex remainder of Lemma 3.1. The two models differ where it matters: the fugacity n^α of `a291698` is constant during coefficient extraction and produces finitely many conjugate resonance sectors (its Theorem 2), whereas $1/(1-q)$ is large only near $q = 1$, and the minor-arc bound (16) leaves no resonance sectors at all. In Lean, the repository has the ingredients `hasSum_jacobi_triple_product` (`Analysis/FabiusFunction/Lean/FabiusFunction/JacobiTripleProduct.lean`) and the eta inversion `Fabius.eta_modularPoint_neg_inv` (`.../FabiusFunction/QPochhammerModularAsymptotic.lean`); neither identity is formalized.

3 The uniform logarithmic expansion

Let $L = \log(1/t)$. Expansion of the exact core gives

$$\log F(e^{-t}) = \frac{L^2/2 + c}{t} - 1 + \sum_{j=1}^J t^j P_j(L) + O_J(t^{J+1}(1+L)), \quad (15)$$

where every P_j is affine in L . In addition to P_1, P_2 above,

$$P_3 = \frac{2L + 245}{5760}, \quad P_4 = -\frac{11}{900}, \quad P_5 = -\frac{8L + 2037}{1451520}, \quad P_6 = \frac{341}{52920}.$$

For a finite coefficient rule, rewrite the j th term of G as

$$\frac{(-1)^{j+1}(1 - e^{-t})^{j-1}}{j(1 + e^{-t} + \dots + e^{-(j-1)t})}.$$

Only $j \leq J + 1$ contributes through degree J . Also $\lambda = L + t/2 - t^2/24 + t^4/2880 - \dots$. In (10), the order-zero terms $L/2$ and $-L/2$ cancel, and the first term of G is exactly one.

For an independent symbolic route, Euler–Maclaurin gives

$$\log F(e^{-t}) \sim -\frac{\text{Li}_2(-e^\lambda)}{t} - \frac{1}{2} \log(1 + e^\lambda) - \sum_{m \geq 1} \frac{B_{2m} t^{2m-1}}{(2m)!} \text{Li}_{2-2m}(-e^\lambda).$$

The reflection identity $-\text{Li}_2(-e^\lambda) = \lambda^2/2 + c + \text{Li}_2(-e^{-\lambda})$ recovers the same coefficients. We use the exact modular identity to justify the complex remainder.

Lemma 3.1. *For a fixed sufficiently large K_0 , uniformly on $|z - t| \leq t/(K_0 L^2)$, formula (15) holds with t replaced by z and its displayed L replaced by $\text{Log}(1/z)$. The remainder is still $O_J(t^{J+1}(1+L))$.*

Proof. For small t , $|\delta(z)| \leq 2t < 1$ and $\Re z \geq t/2$, so

$$|1 - e^{-jz}| \geq 1 - e^{-j\Re z} \geq c_0 \min(jt, 1).$$

The tail of (11) beyond $j = J + 1$ is bounded by $C \sum_{j \geq J+2} (2t)^j / [j \min(jt, 1)] = O_J(t^{J+1})$. Each earlier term has a removable singularity at zero. The function $\lambda(z) - \text{Log}(1/z)$ is analytic there as well; the explicit logarithm accounts for the factor $1 + L$.

Furthermore $\Re(1/z) \sim 1/t$ and $|\Im(\lambda(z)/z)| = O(1/(tL))$. The nonzero theta terms are bounded by

$$\exp\{-c_1 m^2/t + C_1 |m|/(tL)\},$$

whose sum is $O(e^{-c_2/t})$. The dual Euler product obeys a similar bound. Analytic continuation of (9) is valid on this neighborhood: $\delta(z)$ has positive real part, the small-fugacity partner has no zero, and the theta factor is close to one. These observations also specify the logarithm branches. $\square$

The convergent exact j -series is not a convergence assertion for its power-log expansion at zero. Its individual terms have boundary root-of-unity singularities accumulating there. Our power-log statements are finite-truncation asymptotic statements.

[write] Completion of the proof of Lemma 3.1, 5 October 2026. The last paragraph of the proof is condensed: “the dual Euler product obeys a similar bound”, and the branches are named but not fixed. Here are the details; they use nothing beyond the manuscript’s ingredients. Write $z = t(1 + w)$ with $|w| \leq \eta := 1/(K_0 L^2) \leq 1/10$, and use the principal logarithm Log and the bound $|\text{Log}(1 + x)| \leq 2|x|$ for $|x| \leq \frac{1}{2}$.

(a) *Real part.* $\Re(1/z) = t^{-1}(1 + \Re w)/|1 + w|^2 \geq t^{-1}(1 - \eta)/(1 + \eta)^2 \geq 1/(2t)$, and $\Re z \geq t(1 - \eta) \geq t/2$.

(b) *The dual Euler product.* With $Q(z) = e^{-4\pi^2/z}$, (a) gives $|Q| \leq e^{-2\pi^2/t} \leq \frac{1}{2}$. Hence $\ell_Q(z) := \sum_{k \geq 1} \text{Log}(1 - Q^k)$ converges uniformly, with $|\ell_Q| \leq 2 \sum_{k \geq 1} |Q|^k \leq 4e^{-2\pi^2/t}$. It is analytic on the disc, real at $z = t$, and $e^{\ell_Q} = (Q; Q)_\infty$, which therefore has no zero there.

(c) *The theta factor.* Since $\delta(z)/z$ is analytic and close to 1 for small $|z|$, and $\arg z$ is small, $\lambda(z) = \text{Log}(1/z) + \varphi(z)$ with $\varphi(z) = -\text{Log}(\delta(z)/z) = z/2 - z^2/24 + \dots$, a power series with real coefficients. Thus

$$\frac{\lambda(z)}{z} = \frac{L - \text{Log}(1 + w)}{t(1 + w)} + \frac{\varphi(z)}{z}.$$

Here $|\Im(L/(1 + w))| = L|\Im w|/|1 + w|^2 \leq 2L\eta = 2/(K_0 L)$, $|\text{Log}(1 + w)/(1 + w)| \leq 3\eta \leq 1/(K_0 L)$ for $L \geq 3$, and, as $\varphi(z)/z$ is analytic with real Taylor coefficients, $|\Im(\varphi(z)/z)| = O(|\Im z|) = O(t\eta)$. So $|\Im(\lambda(z)/z)| \leq 4/(K_0 t L)$ for small t , which is the bound $O(1/(tL))$ of the proof with an explicit constant. For $u(z) = \lambda(z)/z - 1/2$ and $m \neq 0$,

$$|e^{-2\pi^2 m^2/z} e^{2\pi i m u(z)}| = e^{-2\pi^2 m^2 \Re(1/z) - 2\pi m \Im u(z)} \leq \exp\left\{-\frac{\pi^2 m^2}{t} + \frac{8\pi|m|}{K_0 t L}\right\} \leq e^{-\pi^2 m^2/(2t)}$$

once $L \geq 16/(\pi K_0)$. Hence $|\Theta(u(z), z) - 1| \leq 2 \sum_{m \geq 1} e^{-\pi^2 m/(2t)} \leq 4e^{-\pi^2/(2t)} \leq \frac{1}{2}$, so $\text{Log } \Theta(u(z), z)$ is analytic on the disc, real at $z = t$ (where Θ is real and positive), and bounded by $8e^{-\pi^2/(2t)}$.

(d) *The core.* $|\delta(z)| \leq |z|e^{|z|} \leq 2t < 1$ and $\Re \delta(z) = 1 - e^{-\Re z} \cos \Im z > 0$, so $\lambda(z) = -\text{Log } \delta(z)$ is analytic. For $k \geq 0$, $|\delta(z)e^{-kz}| \leq 2t$, so $G(z) := \sum_{k \geq 0} \text{Log}(1 + \delta(z)e^{-kz})$ converges absolutely, and $e^{G(z)} = (-\delta(z); e^{-z})_\infty$. Expanding each logarithm and summing over k first is legitimate because $\sum_{j \geq 1} |\delta|^j/(j(1 - e^{-j\Re z})) < \infty$; this gives (11) at complex z .

(e) *Identity and branch.* Both sides of (9), with $\lambda(z)$, $u(z)$, $Q(z)$ and $G(z)$ as above, are analytic on the disc and agree on its real diameter, hence on the whole disc. So $F(e^{-z})$ has no zero there, and $K(z) + \text{Log } \Theta(u(z), z) - \ell_Q(z)$ is a continuous logarithm of it that is real at $z = t$; on the connected disc it is the only one, and it is the $\log F(e^{-z})$ of the lemma. By (b) and (c) it equals $K(z) + O(e^{-\pi^2/(2t)})$, and the error is $O(t^{J+1})$ for every J . The expansion of $K(z)$ is the first paragraph of the proof. This completes the proof.

4 Coefficient extraction with the precise remainder

Write $r = e^{-t}$, $q = re^{i\theta}$, $D = |1 - q|$, and $\alpha = (1 - r)/D$. The triangle inequality gives $|F(q)| \leq \prod_{k \geq 1} (1 + r^k/D)$. For $k \leq \lfloor \lambda/t \rfloor$, $w_k = r^k/(1 - r) \geq 1$, and

$$\log(1 + w_k) - \log(1 + \alpha w_k) \geq (1 - \alpha)/2.$$

Since $D^2 = (1 - r)^2 + 2r(1 - \cos \theta)$, uniformly for $|\theta| \leq \pi$,

$$\frac{|F(re^{i\theta})|}{F(r)} \leq \exp \left\{ -c_3 \frac{L}{t} \min \left(\frac{\theta^2}{t^2}, 1 \right) \right\}. \quad (16)$$

At $|\theta| \geq t/(K_0 L^2)$ this is $\exp\{-\Omega(1/(tL^3))\}$, negligible to every power of t . All other roots of unity are suppressed simultaneously.

Put $S(t) = (L^2/2 + c)/t$. Then $-S'(t) = n$ and $(-1)^k S^{(k)}(t) = t^{-k-1} V_k(L)$. Cauchy's formula reads

$$a_n = \frac{e^{nt}}{2\pi} \int_{-\pi}^{\pi} F(e^{-t+i\theta}) e^{-in\theta} d\theta.$$

Set $\epsilon = \sqrt{t}$ and $X = \theta\sqrt{V}/t^{3/2}$, with central cutoff $|X| \leq \epsilon^{-1/4}$. On the intermediate arc up to $|\theta| = t/(K_0 L^2)$, Taylor's theorem gives

$$\Re\{S(t - i\theta) - S(t) - in\theta\} \leq -c_4 X^2.$$

The cubic-to-quadratic ratio is $O(|\theta|/t) = O(L^{-2})$. The remaining logarithmic amplitude changes by $O(t(1+L))$, which cannot affect this bound beyond the central cutoff.

On the central arc, Lemma 3.1 replaces the amplitude by its finite sum through degree J , with multiplicative error $O_J(t^{J+1}(1+L))$. The normalized integrand is $e^{-X^2/2} \exp(Q + R_J)$ with Q, R_J as in Section 5. For a precise remainder, treat ϵ as complex while L and real X are fixed. On the auxiliary circle

$$|\epsilon| = \rho = \frac{c_J}{(1+L)^{1/2}(1+|X|)^3},$$

Q and R_J are uniformly bounded. Indeed, $|w| = |\epsilon X|/\sqrt{V} \ll 1$, $Q = O(|\epsilon| |X|^3/(1+L))$, and $R_J = O_J(|\epsilon|^2(1+L))$. The exact phase has a removable singularity at $\epsilon = 0$ after its linear and quadratic terms have been separated. Cauchy's Taylor remainder through degree $2J+1$ is therefore bounded by

$$C_J |\epsilon|^{2J+2} (1+L)^{J+1} (1+|X|)^{6J+6}.$$

For actual $\epsilon = \sqrt{t}$, $|\epsilon| \leq \rho/2$ on the central cutoff for large L , because $\epsilon^{1/4}(1+L)^{1/2} \rightarrow 0$. Gaussian integration yields exactly $O_J(t^{J+1}(1+L)^{J+1})$. Odd Taylor terms vanish on the symmetric interval. Extending the finitely many even moments to the real line costs an exponentially small error. Expansion through the odd degree before cancellation avoids a half-power loss.

The Jacobian is $t^{3/2}/\sqrt{V}$ and $S(t) + nt = t^{-1}B(L)$. The constant amplitude is e^{-1} . These facts prove Theorem 1.1, including its coefficient growth bound.

5 A finite coefficient rule

Let $\mathcal{G}$ be the Gaussian moment functional $\mathcal{G}(X^{2m}) = (2m-1)!!$, $\mathcal{G}(X^{2m+1}) = 0$. For $w = -i\epsilon X/\sqrt{V}$, set

$$Q = \sum_{k \geq 3} \frac{i^k \epsilon^{k-2} V_k X^k}{k! V^{k/2}},$$

$$R_J = \sum_{r=1}^J \epsilon^{2r} (1+w)^r P_r(L - \log(1+w)).$$

Then for $0 \leq j \leq J$,

$$C_j(L) = \mathcal{G}([\epsilon^{2j}] \exp(Q + R_J)). \quad (17)$$

Only $k \leq 2j+2$ and $r \leq j$ enter. Gaussian parity removes half-powers of t and square roots of V , giving the asserted rational functions. For implementation, if $H(\epsilon) = \sum_{k \geq 1} h_k \epsilon^k$ and $e^H = \sum b_k \epsilon^k$, use

$$b_0 = 1, \quad b_n = \frac{1}{n} \sum_{k=1}^n k h_k b_{n-k}.$$

The included independent symbolic reconstruction verifies (7)–(8), odd cancellation, and computes C_3 .

6 Growth inversion

For a large target Y , put $y = \log Y$ and choose a new L from

$$y = H(L) := e^L B(L). \quad (18)$$

This inverse-carrier coordinate differs from the forward saddle at a preassigned n . Set

$$N_0 = e^{2L} A(L), \quad T = 1 + \frac{3}{2}L + \frac{1}{2} \log(2\pi V), \quad T' = \frac{3}{2} + V'/(2V).$$

Theorem 6.1. *The explicit function*

$$\hat{N}(Y) = N_0 + e^L T + \frac{T^2 + 2TT'}{2V} - C_1(L) \quad (19)$$

satisfies $\hat{N}(a_n) = n + O(e^{-L}(1+L)^3) = n + o(1)$. Nearest-integer rounding therefore recovers the index on the sequence range for every sufficiently large n .

Proof. Write $N(L) = e^{2L} A(L)$. Exactly,

$$H' = e^L V, \quad N' = e^{2L} V, \quad \frac{dN}{dH} = e^L. \quad (20)$$

Define a positive smooth first-correction model $M_1(x)$ by the leading factor in (3) times $1 + tC_1$, where $x = N(L)$. It is eventually increasing since $(\log M_1)'(x) = t(1 + o(1))$. The forward logarithmic error $O(t^2 L^2)$ becomes $O(tL^2)$ under its inverse.

For a direct residual verification, let $H(L) = y$ and set

$$\begin{aligned} \Delta_1 &= tT/V, \\ \Delta_2 &= t^2 \left(\frac{TT'}{V^2} - \frac{(V + V')T^2}{2V^3} - \frac{C_1}{V} \right). \end{aligned}$$

Substituting $\Delta_1 + \Delta_2$ into

$$H(L + \Delta) - T(L + \Delta) + \log\{1 + e^{-(L+\Delta)} C_1(L + \Delta)\} = y$$

leaves residual $O(t^2(1+L)^2)$. Its derivative is asymptotic to $t^{-1}V$, so the root error is $O(t^3)$. Expanding $N(L + \Delta)$ through degree two gives exactly (19), with index error $O(t(1+L)^2)$, proving the stated weaker bound. $\square$

The term $-C_1(L)$ is unbounded of order L and cannot be discarded in an inverse intended to recover integer indices.

For all orders let

$$h(L) = -T + \log \left(1 + \sum_{j \geq 1} e^{-jL} C_j(L) \right), \quad D_y = e^{-L} V^{-1} \partial_L.$$

The formal reversion rule is

$$N(L) + \sum_{k \geq 1} \frac{(-1)^k}{k!} D_y^{k-1} \{e^L h(L)^k\}, \quad H(L) = y. \quad (21)$$

At each exponential order it involves finitely many terms. A finite smooth forward model and the same Taylor-residual proof turn each sufficiently deep truncation into a controlled range inverse. For $D_2 = C_2 - C_1^2/2$, the next correction after (19) is $e^{-L}B_1$, where

$$B_1 = -D_2 - \frac{(TC_1)'}{V} + \frac{1}{6V} \left(\frac{T^3 + 3T^2T'}{V} \right)'.$$

A Lambert starting value for (18) is $L_0 = 2W(\sqrt{y}/2)$, solving $y = e^{L_0}L_0^2$; inverse- L_0 corrections account for the rest of B . Keeping the carrier exact is usually simpler. In particular,

$$\hat{N}(Y) \sim \frac{(\log Y)^2}{2(\log \log Y)^2}.$$

[write] **Status of this section, and the transseries volume, 5 October 2026.** Theorem 6.1 is proved above; its proof gives the index error $O(e^{-L}(1+L)^2)$, and the stated $(1+L)^3$ is weaker, as the proof says. Three statements of the text after it are not proved in the manuscript and are recorded in Section 9: that each deep truncation of (21) is a controlled range inverse (Question 9.1); the formula for B_1 (Question 9.2); and that inverse- L_0 corrections account for the rest of B (Question 9.3, settled there by a [write] proof). The *method* of this section and the next is an instance of results of the companion transseries volume (Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex), which the manuscript does not cite; no novelty is claimed for it here. The Lambert start $L_0 = 2W(\sqrt{y}/2)$ solves $X + 2\log X = \log y$, the dominant block of `p0:thm:lambert-core` with $a = 1$, $b = 2$ on its principal branch ($b > 0$). The rule (21) is the formal reversion of `p0:thm:perturbed-inversion` (core $F = H$, correction $E = h$, $\varepsilon = 1$, operator $\mathcal{D} = D_y$), written in its Lagrange–Bürmann form for the function N of the root, with weight $N'/H' = e^L$ from (20). The bracket and the separation caveat of Section 7 are `p0:thm:staircase` (2), and nearest-integer rounding at $Y = a_n$ is its part (3).

7 Thresholds and scope

The $k = 1$ factor of F is $(1 - q)^{-1}$. The remaining product has nonnegative coefficients, and its $k = 2$ factor has positive coefficients in every degree at least two. Thus $a_n > a_{n-1}$ for all $n \geq 2$. The threshold inverse $N(Y) = \min\{n \geq 2 : a_n \geq Y\}$ is well defined. For sufficiently large Y , it lies between the ceilings of

$$\hat{N}(Y) \pm Ce^{-L}(1+L)^3$$

for an appropriate constant C . This follows from monotonicity of the smooth model and the uniform forward error at neighboring integers. Eventually there are at most two candidate ceilings. A single ceiling is justified only when both bracket endpoints have the same ceiling. At $Y = a_n$, use nearest-integer rounding instead.

Formal inversion, inversion on the sequence range, a threshold staircase, and inversion of an exact continuous interpolation are different objects. This paper treats the first three. It specifies no exact sequence interpolation, effective onset, coefficient-sector amplitudes, or Stokes classification. The normalized modular sectors belong to the generating function itself.

8 Checks and literature scope

The accompanying scripts reconstruct P_1 through P_6 independently from Euler–Maclaurin and the exact small-fugacity series. Gaussian extraction independently reproduces C_1, C_2 and odd cancellation. Exact integer coefficient generation and high-precision evaluation give these selected inverse errors:

n	$N_0(a_n) - n$	$\hat{N}(a_n) - n$
100	-24.644779	-0.050465
1000	-81.976080	-0.015455
10000	-261.371189	-0.005225

The modular identity was checked to at least ninety decimal places at five positive arguments. These are diagnostics, not an effective-onset certificate; the theorems rest on the analytic proof.

The live OEIS entry inspected on 1 October 2026 contains no asymptotic formula. The cited combinatorial papers establish interpretations and generating functions. Classical q-product expansions [5] and a recent uniform q-Pochhammer expansion [6] are relevant background. The latter’s stated uniform corollary imposes a fixed lower bound on the real part of its logarithmic argument, which our moving fugacity leaves. The exact pairing here avoids that issue. A bounded literature search did not locate a published coefficient or inverse-growth theorem for A126348; this does not assert exhaustive absence.

[write] Rechecks and shipped names, 5 October 2026. The OEIS entry, read again on 5 October 2026, still contains no asymptotic formula; its terms are licensed CC BY-SA 4.0, and none are shipped here (the recorded values are the programs’ own computations). The five positive arguments of the modular check are $t = 0.4, 0.7, 1.0, 1.5, 2.0$, with absolute residuals of order 10^{-101} (`data/modular_numeric_results.json`). The “accompanying scripts” are shipped as `code/derive_symbolic.py`, `code/verify_symbolic_independent.py` (which also prints C_3), `code/independent_reviewer_check.py`, `code/verify_numeric.py`, `code/verify_inverse.py`, `code/verify_product.py` and `code/verify_modular.py`; `derive_symbolic.py` recomputes $P_1, \dots, P_6$ but evaluates C_2 from the printed formula (a transcription check), while the two independent programs derive C_1 and C_2 . The reference of Naranjo and Ramírez [3] is also cited by `a022629-distinct-partition-norms`, for the different product $\prod_k (1 + kq^k)$. The report `a271619-strict-twice-partitions`, placed with this one in `36571ae0e`, treats in its member manuscript (bundle Report 181, A358836) the product $\prod_k (1 + q^k/(q; q)_k)$, another distinct-part product with a q -dependent decoration, also reduced by a Jacobi and eta factorization; the two reports share no theorem.

9 Further questions and research

[write] Added 5 October 2026, batch 100. This section is the write’s. Under the repository’s standing rule (4 October 2026) every claim of the manuscript that it states without proof is recorded here as an open question, with its source, the argument sketch it came with and what is missing. No claim of the manuscript was found to be wrong. One sketched step is completed elsewhere: the dual-product bound and the branches in the proof of Lemma 3.1 (the **[write]** note after Section 3). Question 9.3 is settled below. The others are open.

Question 9.1 (All-orders range inverse). *Source:* Section 6, after (21): “A finite smooth forward model and the same Taylor-residual proof turn each sufficiently deep truncation into a controlled range inverse”; `derivation.md` repeats it (“This proves an all-orders range reversion with controlled finite truncations”). *Sketch:* take the positive smooth model given by Theorem 1.1 with $J = K$, note that its logarithm has derivative $t(1 + o(1))$ in $x = N(L)$, and verify the truncation of (21) through exponential order e^{-KL} by a Taylor residual, as the proof of Theorem 6.1 does for $K = 1$. That each order involves finitely many terms is elementary: the k th term of (21) is $e^{-(k-2)L}$ times a function of polynomial growth in L . *Missing:* the residual estimate for a general truncation, and the resulting bound $\hat{N}_K(a_n) = n + O_K(e^{-KL}(1 + L)^{m_K})$ with an explicit exponent m_K .

Question 9.2 (The next inverse correction B_1). *Source:* Section 6: with $D_2 = C_2 - C_1^2/2$ the next correction after (19) is $e^{-L}B_1$, $B_1 = -D_2 - (TC_1)' / V + \frac{1}{6V}((T^3 + 3T^2T')/V)'$. *Sketch*

(`derivation.md`, “One additional inverse order”): keep the first three terms of (21) and the second logarithmic amplitude coefficient; it is called “an optional refinement”. The delivery’s `AUDIT.md` states that B_1 “has the correct signs”. *Evidence only*: at intake (5 October 2026, code independent of the delivery, B_1 with numerical differentiation) adding $e^{-L}B_1$ to $\hat{N}(a_n)$ changed the error at $n = 100, 1000, 10000$ from $-0.050465, -0.015455, -0.005225$ to $0.0088437, 0.00094743, 0.00011451$; the reduction is consistent with an error $O(e^{-2L}(1+L)^m)$. *Missing*: a symbolic derivation of B_1 from (21) and (8), and the range bound for $\hat{N} + e^{-L}B_1$ by the residual argument with $J = 2$; this is the case $K = 2$ of Question 9.1.

Question 9.3 (Corrections to the Lambert start). *Source*: Section 6: $L_0 = 2W(\sqrt{y}/2)$ solves $y = e^{L_0}L_0^2$, and “inverse- L_0 corrections account for the rest of B ”; stated without proof.

Settled at this write ([`write`], 5 October 2026): for large y the root L of (18) has a convergent expansion

$$L = L_0 + \sum_{k \geq 1} d_k L_0^{-k}, \quad d_1 = -1, \quad d_2 = \frac{5}{2} - \frac{\pi^2}{3}, \quad d_3 = \pi^2 - \frac{19}{3}, \quad d_4 = \frac{\pi^4}{18} - \frac{10\pi^2}{3} + \frac{209}{12},$$

with $d_k \in \mathbb{Q}[\pi^2]$. *Proof*. Taking logarithms, $\log y = L + 2 \log L + \log(1 + 1/L + 2c/L^2)$ and $\log y = L_0 + 2 \log L_0$. Subtracting and writing $L = L_0 + \delta$, $s = 1/L_0$, so that $L = L_0(1 + \delta s)$ and $1/L = s/(1 + \delta s)$, the root satisfies $\Gamma(\delta, s) = 0$ with

$$\Gamma(\delta, s) = \delta + 2 \operatorname{Log}(1 + \delta s) + \operatorname{Log}\left(1 + \frac{s}{1 + \delta s} + \frac{2cs^2}{(1 + \delta s)^2}\right).$$

Γ is analytic near $(0, 0)$ with $\Gamma(0, 0) = 0$ and $\partial_\delta \Gamma(0, 0) = 1$, so by the implicit function theorem there are $\rho, \sigma > 0$ and an analytic $\delta(s)$ on $|s| < \sigma$, with $\delta(0) = 0$, such that for $|s| < \sigma$ it is the only zero of $\Gamma(\cdot, s)$ in $|\delta| < \rho$. The actual root lies in that disc for large y : the real function $\delta \mapsto \delta + 2 \log(1 + \delta s)$ has derivative at least 1 for $\delta > -L_0$, and at the root it equals $-\log(1 + 1/L + 2c/L^2) \in (-3/L, 0)$, so $|\delta| \leq 3/L \rightarrow 0$. Hence $\delta = \delta(1/L_0)$, a convergent power series in $1/L_0$; its coefficients follow by substituting $\delta = \sum_k d_k s^k$ into Γ and solving order by order. The values above were computed so with SymPy and checked against the exact root: at $y = 10^{12}$ ($L_0 = 21.495 \dots$) the five-term sum (through d_4) leaves an error of 4.3×10^{-6} , of the size of $d_5 L_0^{-5}$, where $d_5 = \frac{35\pi^2}{3} - \frac{4\pi^4}{9} - \frac{763}{15} \approx 20.99$; adding d_5 lowers it to 3×10^{-7} . (Dated note, 5 October 2026: the first version of this sentence gave 3×10^{-7} for the five-term sum; an independent check made by the intake after the write found that figure belongs to the six-term sum, recomputed $d_1, \dots, d_4$ exactly and found the implicit-function argument valid.) $\square$

Question 9.4 (Effective onset). *Source*: Sections 7 and 8 and the delivery’s status line: all statements are eventual, with no explicit n_0 or constants; the numerical table is “diagnostics, not an effective-onset certificate”. *Missing*: explicit constants in Lemma 3.1, in the minor-arc bound (16) (the constants c_0, c_3), in the intermediate-arc and Cauchy-circle estimates of Section 4, and an interval-arithmetic check below the resulting onset. With them, the nearest-integer rule of Theorem 6.1 and the threshold bracket of Section 7 would become finite-input certificates.

Question 9.5 (Exponentially small coefficient sectors and Stokes data). *Source*: after Theorem 2.1 and in Section 7: the sectors $D_r(u)e^{-2\pi^2 r/t}$ belong to the generating function and “do not by themselves identify exponentially small coefficient sectors”; no coefficient-sector amplitudes and no Stokes classification are given. *Sketch available*: the minor-arc bound proves that the remainder is beyond every fixed power of t , and (13) gives the generating-function sectors exactly. *Missing*: a transfer of the sectors $r \geq 1$ through the saddle integral (the phase u moves with t), the resulting coefficient-level amplitudes, and their Stokes behaviour. Whether such sectors are visible against the algebraic remainder of Theorem 1.1 at all is part of the question.

Question 9.6 (An exact interpolation). *Source:* Section 7: formal inversion, range inversion, the threshold staircase and inversion of an exact continuous interpolation are different objects, and only the first three are treated; no exact sequence interpolation is specified (also `derivation.md`, “Inversion on the sequence range”). *Context:* by `p0:thm:staircase` (1) of the transseries volume, the threshold equals $\lceil \nu \rceil$ for *every* admissible interpolation, so the integer results do not depend on a choice. *Missing:* a canonical real-analytic interpolation of (a_n) (for instance from the saddle integral at real n), its monotonicity, and its inverse.

References

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